

Composite screening report

Educational CLT and cylinder screening. Reference data, not qualified design allowables.

Inputs and load cases

Material: T300/5208. 16 physical plies; thickness 2 mm.
Bottom to top: 0°; 45°; -45°; 90°; 0°; 45°; -45°; 90°; 0°; 45°; -45°; 90°; 0°; 45°; -45°; 90°
Each ply: 0.125 mm, M0.
M0: Sidebar material
Laminate loads: N [kN/m] = (100, 0, 0); M [N·m/m] = (0, 0, 0).

Laminate stiffness and response (sidebar mechanical loads)

A [MN/m]	B [N]	D [N·m]
152.7 / 45.21 / 0	-1.607e+04 / 0 / -2679	51.53 / 14.46 / 0
45.21 / 152.7 / 0	0 / 1.607e+04 / -2679	14.46 / 51.53 / 0
0 / 0 / 53.76	-2679 / -2679 / 0	0 / 0 / 17.3

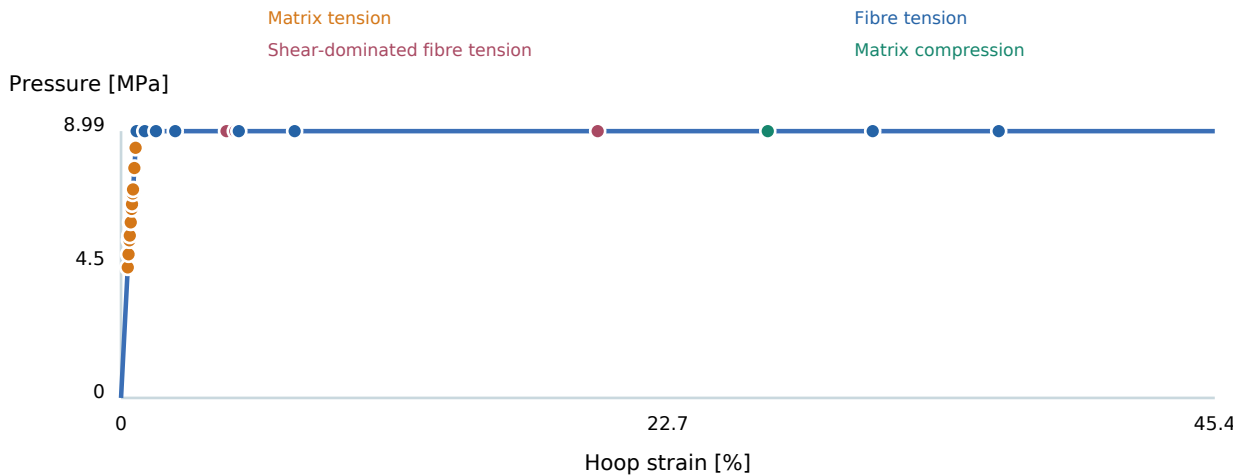
Each matrix row uses x, y, xy order. E_x / E_y / G_{xy} [GPa]: 67.06 / 67.06 / 26.77.
Mid-plane strain [μϵ]: 745.6 / -219.7 / 11.8; curvature [1/m]: 0.2321 / 0.00405 / 0.08142.
Mechanical first-ply load factor: 4.036 (Tsai-Wu), ply 16, Top face. Entered loads are below or at the initiation limit.

Cylinder pressure case (separate from sidebar loads)

R = 100 mm; current layup h = 2 mm; working pressure = 10 MPa. Nx=pR/2, Ny=pR.

First-ply CLT [MPa]	First-ply Hashin [MPa]	Last-ply model stop [MPa]	Netting [MPa]
4.4068	4.3990	8.9907	15.0000

First-ply marks calculated initiation. Last-ply is the assumed algorithm stop, not an ultimate or burst load. Netting is a separate fibre-only equilibrium reference.



Points: initiation modes before stiffness reduction; jumps: redistribution at held pressure. This is not a stability analysis.

Limits and validation

Not validated against experiment. No like-for-like comparison is possible with the current cases: inner radius is not explicitly reported in the supplied research text; Type III liner load sharing is not modelled. Sources remain unverified against the original paper pages. See validation/README.md.
Model assumptions: plane stress, perfect bonding, thin-plate CLT; Tsai-Wu interaction F12 is assumed. Cylinder degradation: E1 retained 0.01; E2 / G12 retained 0.1 / 0.1; Hashin transverse shear strength assumed when not supplied. Not included in this export: thermal preload, dome strength, liner/boss load sharing, interlaminar damage, fatigue, leakage, buckling, manufacturing defects or temperature-dependent properties.